Do Patient Metadata Still Help Strong Image Models? Representation-Dependent Attenuation of Auxiliary Information in Skin Lesion Classification

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Abstract—Fusing patient metadata with lesion images is widely reported to improve skin cancer classification. We ask whether this benefit is a property of the data or of a weak image model. We define the added value of metadata as the out-of-fold reduction in cross-entropy when metadata are added to an image score, and relate its population counterpart, $I(Y; A \mid S)$, exactly to information the image cannot carry plus a representation-dependent residual. In a pre-registered analysis on HAM10000, added value fell from 18.6 to 1.7×10^{-3} nats from colour histograms to ConvNeXt-T. In two pre-registered replication waves covering 6 further datasets, this attenuation held in all 6 after Holm correction and survived a strictly cross-fitted estimator, complete cases and every missing-data encoding, whereas attenuation between ResNet-18 and ConvNeXt-T depended on the analysis. An exploratory intervention locates the mechanism: erasing the linear trace of age from the image representation raised age's added value from 2.0 to 13.2×10^{-3} nats on HAM10000, beyond performance-matched random erasure. The remaining benefit was concentrated: the quarter of lesions with the largest expected gain carried at least 70% of it, whereas a pre-registered nearest-neighbour graph of the representation added no predictive information. The non-redundant share of the metadata's information ranged from 0.01 to 0.50 and depended on the label definition. We also document three evaluation pitfalls: stacking with shared folds leaks test labels, missing-value encoding can halve or triple added value, and circular-shift saliency tests are invalid for centred lesions.

Index Terms—skin lesion classification, patient metadata, multimodal fusion, conditional mutual information, representation learning, pre-registration, cross-fitting, concept erasure

I. Introduction

Clinicians do not diagnose skin lesions from images alone: age, anatomical site, and lesion history all shape the assessment. Following this intuition, many automated systems fuse patient metadata with image features. A systematic review of such systems found that every included study reported a gain from patient data, while noting that it remains unclear how each variable contributes [1]. On the smartphone dataset PAD-UFES-20, adding clinical data improved balanced accuracy by about 7% across several networks [2].

A reported gain, however, conflates two things: information in the metadata that the image genuinely lacks, and information the image contains but a particular model failed to extract. A patient's age, for example, is partly visible in the skin around a lesion. If an image model is weak, metadata can compensate for what it misses; if it is strong, the same metadata may be redundant. Whether metadata "help" may therefore depend on the image model they are added to, a dependence that single-backbone studies cannot reveal.

This paper makes five contributions:

1) An interpretation of added value. Using the chain rule of mutual information, we write the Bayes added value of metadata A to a representation $S = g(X)$ of an image X as $I(Y; A \mid S) = I(Y; A \mid X) + R(S)$, where $I(Y; A \mid X)$ is the information unavailable from the image and the residual $R(S)$ vanishes as $S \to X$ (Proposition 1). The decomposition is elementary; its value is that it says what a measured gain from metadata does and does not show. It does not imply that added value falls as representations improve; it does predict that removing from S what it shares with A restores A's value, which we test by intervention.
2) A measurement protocol. We estimate added value out of fold along a ladder of frozen image representations, with grouped cross-validation, inner-fold tuning, strict cross-fitting of the two stages, and grouped bootstrap inference, and we audit metadata for label leakage, including label-informative missingness.
3) Pre-registered evidence on seven datasets. A pre-registered analysis on HAM10000 [3] and two pre-registered replication waves: ISIC 2020 [4], ISIC 2019 excluding HAM10000 [5] and PAD-UFES-20 [6]; then HIBA [7], MSK 2020 [8] and MILK10k [9], whose paired dermoscopic and clinical images of the same lesions allow a test of imaging modality. Two pre-registered mechanism tests failed, and we report why.
4) Interventions on the representation. Fine-tuning, training-set size, backbone size within a family, a 304M-parameter vision transformer, and linear

concept erasure separate what attenuates metadata value (a better image model for the task; image features that encode the variable) from what does not (more parameters on frozen features).

5) Where metadata help, and evaluation pitfalls. Benefit concentrates in lesions the image model finds uncertain and can be ranked without labels, whereas a representation graph adds nothing beyond the fitted models (pre-registered). Stacking with shared folds leaks test labels, missing-value encoding and image- vs. patient-weighting can change conclusions, and the circular-shift null used for saliency–mask overlap is invalid when lesions are centred.

II. Related Work

Metadata fusion in dermatology. Concatenating one-hot metadata with image features is the dominant approach, and all eleven studies in a systematic review reported a gain [1]. The closest prior evidence is a benchmark of nine backbones on an ISIC subset [10]: averaged over its backbones, metadata raised AUROC by 0.023 for models trained from scratch but by only 0.001 for ImageNet-pretrained, fine-tuned models (our calculation from its Table 5; single runs, no intervals). The authors did not frame this as attenuation, and to our knowledge no study has measured added value along a controlled ladder of image models with grouped inference and pre-specified tests. A reader study on MILK10k found dermoscopy and clinical close-ups diagnostically complementary [11]; whether the modality changes what metadata add to an automated model has not been tested.

Incremental value. In clinical prediction, the added value of a new variable is commonly tested with nested models; comparing nested-model AUCs with DeLong's test is invalid [12]. In logistic models, omitting a predictor that is independent of a covariate still attenuates that covariate's coefficient (non-collapsibility) [13]. This mechanism makes directly acting metadata gain value as the image model improves (Section III).

Per-sample utility and concept erasure. The expected per-sample reduction in log-loss from observing a feature equals its pointwise conditional mutual information, and dynamic feature selection trains networks to predict it [14], [15]; we use the same quantity to ask where metadata help, not to select features. Linear concept erasure removes all linearly decodable information about a concept from a representation in closed form [16]; we use it as an intervention on the image representation.

Leakage and shortcuts. Splitting medical images without grouping by subject inflates accuracy substantially [17]. Dermatology datasets contain several images per lesion and per patient, and acquisition source can predict diagnosis.

Spatial nulls. Comparisons between spatial maps must preserve spatial autocorrelation in the null distribution; naive voxel permutation is anti-conservative [18], [19]. We carry this principle over to the evaluation of saliency maps against anatomy.

III. Theory

Let X be an image, A metadata, $Y \in \{0,1\}$ the label, and $S = g(X)$ a representation. With log-loss $\ell(y,p) = -\log p(y)$, the Bayes predictors satisfy $\mathbb{E}\,\ell(Y, P(Y \mid S)) = H(Y \mid S)$, so the expected reduction in log-loss from adding A is

$$\Delta(S) = H(Y \mid S) - H(Y \mid S, A) = I(Y; A \mid S). \tag{1}$$

Proposition 1. For any measurable g and $S = g(X)$,

$$I(Y; A \mid S) = I(Y; A \mid X) + \underbrace{I(Y; X \mid S) - I(Y; X \mid S, A)}_{R(S)}. \tag{2}$$

If g is injective with a measurable inverse, $R(S) = 0$. If moreover $Y \perp\!\!\!\perp A \mid X$, then $\Delta(S) = 0$.

Proof. Expand $I(Y; A, X \mid S)$ with the chain rule in two orders [20]: $I(Y; A \mid S) + I(Y; X \mid S, A) = I(Y; X \mid S) + I(Y; A \mid X, S)$. Because S is a function of X, conditioning on (X, S) equals conditioning on X, so $I(Y; A \mid X, S) = I(Y; A \mid X)$. Rearranging gives (2). If g is injective, X is a function of S, so both terms of $R(S)$ vanish. The final claim follows because $I(Y; A \mid X) = 0$ when $Y \perp\!\!\!\perp A \mid X$. □

Interpretation. $I(Y; A \mid X)$ is the part of the metadata's information that no image model can recover. $R(S)$ measures how the representation's losses interact with A. $R(S) > 0$ when A substitutes for image information that S discarded, and $R(S) < 0$ indicates synergy, the information-theoretic face of non-collapsibility [13]. The identity does not order representations: for nested $S_1 = h(S_2)$, $\Delta(S_1) - \Delta(S_2) = R(S_1) - R(S_2)$ can take either sign (if $X = A = Y$ is one fair bit, refining S from a constant to X removes one bit of added value; if $Y = X \oplus A$ for independent fair bits, it creates one), and the frozen representations we compare are not nested. What it does predict is the effect of an intervention: removing from S the information it shares with A should raise Δ when that shared information is what made A redundant (Section V-H). Injective Borel maps between standard Borel spaces have measurable inverses, which covers images and feature vectors. Proposition 1 was checked numerically to machine precision on 200 random discrete distributions and coarsenings (code release).

Estimation. We estimate Δ for each rung k of a capacity ladder by a two-stage out-of-fold procedure. First, an image score $s_k(x)$ is obtained from frozen features $f_k(x)$ by L_2-regularised logistic regression, with the penalty chosen by grouped inner cross-validation within each training fold. Second, logistic models on $[s_k]$ and $[s_k, a]$ are fitted and evaluated out of fold:

$$\hat{\Delta}_k = \frac{1}{n} \sum_{i=1}^{n} \Big[\ell\big(y_i, \hat{p}_0^{(-i)}\big) - \ell\big(y_i, \hat{p}_1^{(-i)}\big) \Big], \tag{3}$$

where $(-i)$ denotes a model fitted without the fold containing i. In the registered analyses the second stage was trained on the out-of-fold scores of the other folds, which came from first-stage models that had seen the test fold; Section IV describes a strictly cross-fitted version, used for every analysis added after the audit. For fixed fitted models q_0, q_1, the expected held-out reduction equals $I(Y; A \mid s_k) + \mathrm{KL}(p_{Y|s} \,\|\, q_0) - \mathrm{KL}(p_{Y|s,A} \,\|\, q_1)$, so $\hat{\Delta}_k$ estimates conditional mutual information only for well-specified second stages; its target conditions on the one-dimensional score s_k, which agrees with $I(Y; A \mid S_k)$ only if s_k is sufficient for Y given S_k with and without A; and it can be negative in finite samples. $\hat{\Delta}_k$ averages over images, so lesions or patients with many images weigh more. Uncertainty is quantified by a bootstrap over grouping units (lesions or patients) with the fitted models held fixed, which omits training variability [21], and differences between rungs by a paired grouped bootstrap of per-image loss differences.

Where metadata help. With $u(s) = I(Y; A \mid S = s)$, $\Delta(S) = \mathbb{E}\, u(S)$ and $u(s) \le H(Y \mid S = s)$: metadata can help only where the image leaves the label uncertain. For a calibrated fusion model, the expected per-image benefit given (s, a) is $K = \mathrm{KL}(q_1 \| q_0)$, which needs no label [15]. Writing $\Delta(S) = H(Y \mid S)\,\eta(S)$ with $\eta \in [0, 1]$ splits attenuation into the image resolving more cases (H falls) and metadata resolving a smaller share of what remains (η falls).

IV. Experimental Setup

Datasets (Table I). HAM10000 (dermoscopy; 10,015 images of 7,470 lesions) [3]; ISIC 2020 (dermoscopy; 33,126 images of 2,056 patients) [4]; ISIC 2019 with all HAM10000 lesions removed [5]; PAD-UFES-20 (smartphone images; 2,298 images of 1,373 patients) [6]; the dermoscopic images of HIBA [7]; MSK 2020, consecutive biopsies from one centre [8]; and MILK10k, in which every lesion has one dermoscopic and one clinical close-up image [9]. The two MILK10k image sets are analysed separately but are not independent. Among datasets that carry ISIC identifiers, no image appears in two datasets; PAD-UFES-20 has no such identifiers.

Leakage audit. Fields that encode how the diagnosis was established were excluded: HAM10000's dx_type (alone, AUROC 0.76 for melanoma) and PAD-UFES-20's biopsed (every cancer is biopsied). On HAM10000 the image source alone predicts melanoma (AUROC 0.73), and frozen deep features identify the source almost perfectly (AUROC up to 0.995), so we report source-adjusted analyses. A separate audit of missingness found that it, too, can carry the label. In PAD-UFES-20, all 804 images lacking gender, skin type, and history fields are benign and unbiopsied, so missingness is a proxy for the excluded biopsed field, and "unknown" growth and change histories are 88% malignant. Smaller associations exist elsewhere (e.g. ISIC 2020: no melanoma among the 65 images of unknown sex). As a further control, we refit the second stage after permuting the metadata between grouping units (50 permutations). Because the pre-registered encoding keeps missing-value indicators, we add a complete-case sensitivity analysis (Section V-F).

TABLE I

Datasets. Positives: melanoma for HAM10000 and ISIC; malignant otherwise. Groups define the cross-validation and bootstrap units. *Images without a lesion identifier form their own group. †No lesion identifier is available. The two MILK10k rows show the same lesions.

Dataset	Modality	Images	Groups	Pos.	Metadata
HAM10000	dermoscopy	10,015	7,470 lesions	1,113	age, sex, site
ISIC 2020	dermoscopy	33,126	2,056 patients	584	age, sex, site
ISIC 2019 (non-HAM)	dermoscopy	15,316	6,461 lesions*	3,409	age, sex, site
PAD-UFES-20	smartphone	2,298	1,373 patients	1,089	demogr. + symptoms
HIBA	dermoscopy	1,224	563 patients	533	age, sex, site, skin type
MSK 2020	mixed	1,174	1,174 images†	148	age, sex, site
MILK10k (derm.)	dermoscopy	5,117	5,117 lesions	3,634	age, sex, site
MILK10k (clin.)	clin. close-up	5,117	5,117 lesions	3,634	age, sex, site

Ladder. HSV colour histograms (48-d); ImageNet-pretrained ResNet-18 and ResNet-50 [22], ConvNeXt-T [23], and ViT-B/16 [24]; frozen, pooled features from 224×224 centre crops.

Protocol. 5-fold stratified grouped cross-validation (lesion or patient) with three fold assignments; inner grouped 3-fold CV for the penalty; 2,000 grouped bootstrap resamples. The HAM10000 analysis and both replication waves were pre-registered, with hypotheses, the Holm correction [25] and its family (six tests in wave 1, eight in wave 2) fixed before the respective data were modelled. Wave 2 restricted HIBA to dermoscopic images in a dated addendum written after reading its metadata and before any modelling. Additions, all exploratory and labelled as such: the complete-case analysis (prompted by the missingness audit), the retained fraction ρ, the source-adjusted and paired HAM10000 analyses, the interpretability analyses, the representation-strength manipulations, and the recoverability, permutation, and cross-dataset analyses; the last group was described in a dated registry before implementation. Two implementation errors found during the work (an image-name mismatch that dropped ISIC 2019 images, and a missing-value check that missed NaN) were fixed before any affected result was used. A later audit found that the second stage had been trained on scores from first-stage models that saw the outer test fold. The strictly cross-fitted estimator trains both second-stage models on inner grouped out-of-fold scores computed within each training fold, scores the test fold with the first stage fitted on the whole training fold, and imputes metadata with training-fold statistics. Called unchanged, the original code reproduced all 120 stored results; the corrected estimator changed them by at most 0.42 of a bootstrap CI half-width (HAM10000, ConvNeXt-T: 1.7 to 2.4×10^{-3} nats). Registered results are reported as computed; every analysis added after the audit (E40–E49) uses the corrected estimator and was described in a dated registry before implementation.

TABLE II

HAM10000, melanoma vs. rest, lesion-grouped 5-fold CV. AUROC and $\hat{\Delta}$ are means over three fold assignments; the CI is the 95% lesion bootstrap for assignment 0. Recoverability: out-of-fold R^2 for age and mean AUROC for the five commonest sites and for image source, predicted from the frozen features.

Representation	Image AUROC	$\hat{\Delta}$ (10^{-3} nats)	95% CI (10^{-3})	Recoverability age	site	source
HSV histogram	0.765	18.6	[12.2, 24.3]	0.06	0.65	0.921
ResNet-18	0.853	3.7	[−0.1, 8.1]	0.28	0.73	0.979
ResNet-50	0.850	3.0	[−0.5, 6.9]	0.32	0.73	0.989
ConvNeXt-T	0.889	1.7	[−2.7, 5.6]	0.39	0.79	0.995
ViT-B/16	0.887	2.2	[−1.1, 6.3]	0.37	0.79	0.995

TABLE III

HAM10000 robustness (fold assignment 0). ΔAUROC and ΔBrier (10^{-3}) from adding metadata, with 95% lesion-bootstrap CIs. Last two columns: out-of-fold log-loss of score + metadata (two-stage, as in Table II) vs. all frozen features + metadata in one model; similar values mean the one-dimensional score loses little.

Representation	ΔAUROC [95% CI]	ΔBrier [95% CI]	Log-loss with metadata two-stage	joint
HSV histogram	+0.043 [+0.030, +0.056]	4.84 [3.02, 6.70]	0.288	0.285
ResNet-18	+0.007 [+0.001, +0.013]	1.29 [0.23, 2.42]	0.253	0.250
ResNet-50	+0.005 [−0.001, +0.011]	1.16 [0.14, 2.23]	0.255	0.255
ConvNeXt-T	+0.003 [−0.002, +0.007]	0.88 [−0.14, 1.93]	0.231	0.228
ViT-B/16	+0.004 [−0.001, +0.008]	0.79 [−0.25, 1.86]	0.230	0.228

V. Results

A. Pre-registered analysis on HAM10000

Fig. 1 and Table II show that metadata reduce out-of-fold log-loss by 18.6×10^{-3} nats for colour histograms, with a CI excluding zero in all three fold assignments. For the deep representations the reduction is only 3.7×10^{-3} to 1.7×10^{-3} nats, and the log-loss CIs include zero. Table III expresses the same effect in familiar units: the AUROC gain falls from 0.043 [0.030, 0.056] for colour histograms to 0.007 [0.001, 0.013] for ResNet-18, which remains detectable, and to 0.003 [−0.002, 0.007] for ConvNeXt-T, which does not. Fitting all frozen features and metadata in one model gives nearly the same log-loss as the two-stage design (Table III), so compressing the image to a one-dimensional score does not hide an interaction.

Recoverability of age from the image rises along the ladder (R^2 from 0.06 to 0.39), consistent with Proposition 1 for metadata whose information the image carries. A per-variable analysis did not sharpen this: age, sex, and site retained 14%, 22%, and 18% of their colour-histogram added value at the strongest rungs, and even sex proved recoverable from dermoscopy (AUROC 0.77). Because all three variables became recoverable together, HAM10000 alone cannot isolate the mechanism, which motivates the symptom analysis on PAD-UFES-20 below. Three of four pre-registered predictions held; the predicted non-zero floor was not observed.

B. Shortcut control and paired tests

With image source included in both models, added value still falls (from 15.3×10^{-3} to 2.1×10^{-3} nats). Within single sources, melanoma AUROC rises with capacity, so the deep representations do read lesions and do not merely detect the source. The paired attenuation from ResNet-18 to ConvNeXt-T is 2.4×10^{-3} nats (95% CI $[0.7, 4.0] \times 10^{-3}$). Across all 8 image sets, the real metadata outperformed every one of 50 group-level permutations at ConvNeXt-T on complete cases (empirical $p = 0.02$, the smallest attainable), so the residual added value is not an artefact of fitting extra parameters.

TABLE IV

HAM10000: paired differences in added value between representations, $\hat{\Delta}_a - \hat{\Delta}_b$ (10^{-3} nats, 95% paired lesion-bootstrap CI), fold assignment 0. Exploratory: not corrected for multiplicity.

Comparison	Unadjusted	Source-adjusted
HSV histogram − ResNet-18	14.3 [10.7, 18.0]	10.6 [7.9, 13.7]
HSV histogram − ConvNeXt-T	16.7 [12.6, 20.6]	13.2 [9.6, 16.7]
ResNet-18 − ConvNeXt-T	2.4 [0.7, 4.0]	2.6 [0.9, 4.0]
ResNet-18 − ViT-B/16	1.4 [−0.1, 2.9]	1.5 [−0.1, 3.0]
ResNet-50 − ConvNeXt-T	1.6 [−0.1, 3.2]	1.8 [0.2, 3.4]

C. Pre-registered replication

Table V and Fig. 1 report both waves. After Holm correction within each registered family, H1 (colour histogram vs. ConvNeXt-T) held in 3 of 3 datasets in wave 1 and 4 of 4 image sets in wave 2, that is, in all 6 independent datasets (MILK10k counted once, and as holding only if both of its image sets held). H2 (ResNet-18 vs. ConvNeXt-T) held in 1 of 3 and 3 of 4 (MSK 2020: $p_{\text{Holm}} = 0.086$), i.e. in 3 of 6 independent datasets. With the strictly cross-fitted estimator, H1 held in 6 and H2 in 4 of 6: ISIC 2020's H2 moved from $p_{\text{Holm}} = 0.092$ to 0.028 on a difference of 0.36×10^{-3} nats. Across fold assignments, patient weighting, alternative label definitions, complete cases and missing-data encodings, H1 failed only where the metadata carried almost no information (MILK10k, melanoma vs. rest: 1.8×10^{-3} nats), whereas H2 depended on the estimator, the label, the fold assignment and the weighting (Appendix F). At ConvNeXt-T the remaining added value (10^{-3} nats, 95% CI) is: ISIC 2020 1.0 [0.3, 1.7]; ISIC 2019 (non-HAM) 9.1 [3.1, 15.3]; PAD-UFES-20 206.4 [179.9, 232.9]; HIBA 2.6 [−13.5, 17.9]; MSK 2020 18.0 [7.2, 28.6]; MILK10k (derm.) 8.7 [0.8, 15.3]; MILK10k (clin.) 5.6 [−2.7, 12.2].

Mechanism test 1 (PAD-UFES-20, wave 1). We predicted that symptoms, being invisible in photographs, would keep their value while demographics attenuated. From HSV to ConvNeXt-T, the demographic block retained 0.506 of its added value and the symptom block 0.474, so neither M1 nor M2 held. The premise was false: even within a diagnostic class, ConvNeXt-T features predict the symptoms with AUROC 0.64 to 0.76 (bleeding and elevation are visible).

D. Pre-registered modality and lesion-size tests

Mechanism test 2 (P1, MILK10k). If metadata substitute for image detail, clinical close-ups, which show

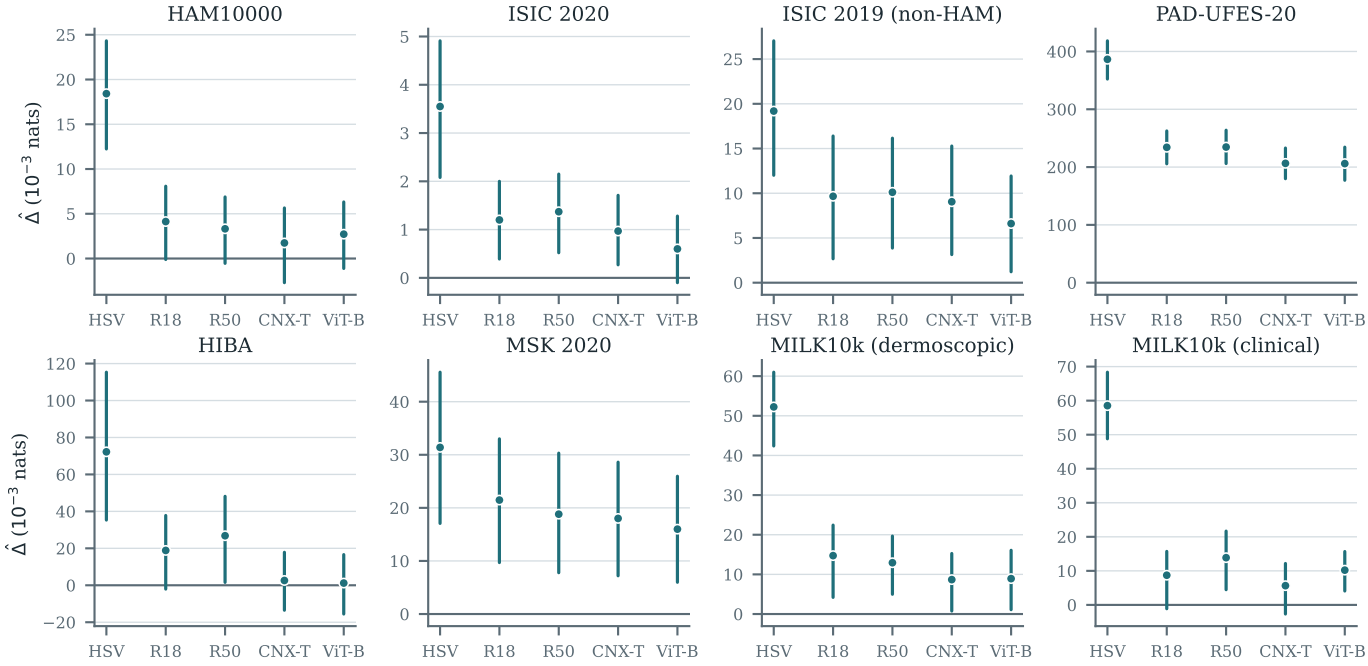


Fig. 1. Added value of patient metadata $\hat{\Delta}$ (out-of-fold log-loss reduction, 10^{-3} nats) along the ladder for all eight image sets, with 95% grouped-bootstrap CIs (fold assignment 0, all images). Panels have different y-axis scales: PAD-UFES-20's values are about two orders of magnitude larger. HSV: colour histogram; R18/R50: ResNet-18/50; CNX-T: ConvNeXt-T; ViT-B: ViT-B/16.

TABLE V
Pre-registered replication in two waves: added value of all metadata $\hat{\Delta}$ (10^{-3} nats, fold assignment 0) along the ladder, and Holm-adjusted one-sided p-values for H1 ($\Delta_{\text{HSV}} > \Delta_{\text{ConvNeXt}}$) and H2 ($\Delta_{\text{ResNet-18}} > \Delta_{\text{ConvNeXt}}$), grouped paired bootstrap. Holm families as registered: 6 tests (wave 1) and 8 tests (wave 2). The two MILK10k rows show the same lesions and are not independent.

Wave	Dataset	HSV	ResNet-18	ResNet-50	ConvNeXt-T	ViT-B/16	H1 p_{Holm}	H2 p_{Holm}
1	ISIC 2020	3.6	1.2	1.4	1.0	0.6	0.003	0.092
1	ISIC 2019 (non-HAM)	19.2	9.7	10.1	9.1	6.6	0.003	0.241
1	PAD-UFES-20	386.4	234.2	234.6	206.4	206.1	0.003	0.003
2	HIBA	72.2	18.9	26.8	2.6	1.2	0.004	0.004
2	MSK 2020	31.4	21.5	18.8	18.0	16.0	0.004	0.086
2	MILK10k (derm.)	52.2	14.7	12.9	8.7	8.9	0.004	0.004
2	MILK10k (clin.)	58.5	8.7	13.9	5.6	10.2	0.004	0.004

less detail than dermoscopy, should leave more of the metadata's information non-redundant. On the same 5,117 lesions the opposite held: $\rho = 0.069$ for clinical vs. 0.105 for dermoscopic images at ConvNeXt-T, a paired difference of -0.036 [-0.071, -0.004]. P1 is therefore rejected. A plausible reason is that its premise again failed in our setting: the two image types gave almost identical image-only AUROC (0.842 vs. 0.845), so the clinical representation was not weaker, consistent with a reader study that found the two modalities complementary rather than ranked [11].

P2 (MSK 2020). Measured lesion diameter added 8.6×10^{-3} nats [1.3, 16.4] to the colour-histogram model but only 1.8×10^{-3} [-1.7, 5.1] to ConvNeXt-T, a retained share of 0.21 (registered threshold < 0.5). P2 was registered as exploratory because MSK 2020 is small; the result is consistent with the finding below that deep image scores track lesion size.

E. How much information remains?

Normalising $\hat{\Delta}$ by the metadata's standalone information $\hat{I}_A$ gives the retained fraction ρ (Table VI, Fig. 2). It falls steeply from colour histograms to the deep representations everywhere, but its level at ConvNeXt-T varies from 0.01 (HIBA) to 0.50 (PAD-UFES-20). Dermoscopic datasets lie at both ends of this range, so the floor is a property of the dataset, its label definition and its metadata, not of the imaging modality. The label matters directly: on ISIC 2019, ρ is 0.48 for melanoma but 0.20 [0.14, 0.25] for malignant vs. benign (corrected estimator). Where added value is small, weighting and encoding matter too: HIBA's value at ConvNeXt-T is 3.8 per image but -0.8 per patient, and 12.7 without missing-value indicators (10^{-3} nats).

TABLE VI

Exploratory: retained fraction $\rho_k = \hat{\Delta}_k / \hat{I}_A$ of the metadata's standalone information that is not redundant with image representation k (all images, pre-registered encoding). $\hat{I}_A$: out-of-fold log-loss reduction of a metadata-only model over prevalence (10^{-3} nats). 95% grouped-bootstrap CIs, fold assignment 0. ρ_k can exceed 1 under synergy and fall below 0 in finite samples.

Dataset	$\hat{I}_A$	HSV	ResNet-18	ConvNeXt-T	ViT-B/16
HAM10000	25.9	0.71 [0.60, 0.80]	0.16 [0.00, 0.27]	0.07 [−0.12, 0.20]	0.10 [−0.03, 0.21]
ISIC 2020	4.3	0.83 [0.72, 0.95]	0.28 [0.15, 0.38]	0.23 [0.08, 0.33]	0.14 [−0.04, 0.25]
ISIC 2019 (non-HAM)	20.4	0.94 [0.84, 1.04]	0.47 [0.25, 0.66]	0.44 [0.23, 0.62]	0.33 [0.12, 0.48]
PAD-UFES-20	416.3	0.93 [0.90, 0.95]	0.56 [0.51, 0.61]	0.50 [0.45, 0.54]	0.49 [0.45, 0.54]
HIBA	217.4	0.33 [0.18, 0.51]	0.09 [−0.01, 0.16]	0.01 [−0.08, 0.07]	0.01 [−0.09, 0.07]
MSK 2020	38.7	0.81 [0.68, 0.92]	0.56 [0.39, 0.69]	0.47 [0.29, 0.61]	0.41 [0.22, 0.57]
MILK10k (derm.)	82.3	0.63 [0.58, 0.68]	0.18 [0.06, 0.26]	0.10 [0.00, 0.17]	0.11 [0.00, 0.18]
MILK10k (clin.)	82.3	0.71 [0.66, 0.76]	0.11 [−0.01, 0.18]	0.07 [−0.04, 0.14]	0.12 [0.05, 0.17]

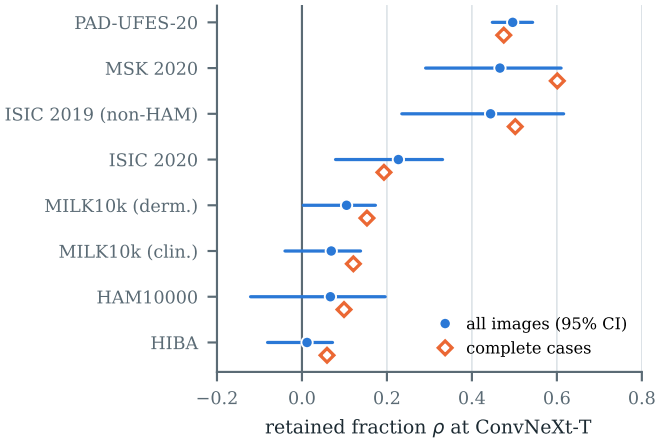


Fig. 2. Retained fraction ρ of the metadata's standalone information left non-redundant by ConvNeXt-T, per image set. Circles: all images with 95% grouped-bootstrap CI (Table VI); diamonds: complete cases (Table VII).

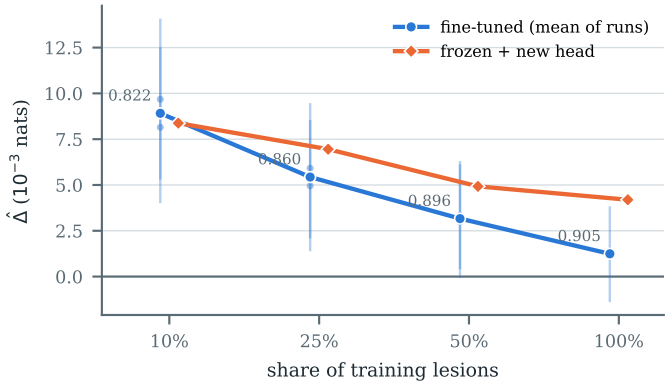


Fig. 3. Exploratory dose–response on HAM10000 (ResNet-18): added value of metadata vs. the share of training lesions available to the image model. All runs on complete cases. Blue: fine-tuned end to end; line through the mean of runs (two subsamples below 100%), small dots and bars are individual runs with 95% CIs, numbers are mean image AUROC. Orange: frozen ImageNet features with a retrained head (mean of two subsamples).

F. Complete-case sensitivity analysis

Restricting every dataset to images with all metadata recorded (Table VII) removes label-informative missingness. H1 holds (uncorrected) in 8 of 8 image sets and H2 in 5 (not in ISIC 2019 (non-HAM), PAD-UFES-20, MSK 2020). On PAD-UFES-20, the standalone information in the metadata falls from 416×10^{-3} to 155×10^{-3} nats, so much of that dataset's apparent metadata signal reflects how missing values arise; its complete cases (1,089 images, mostly biopsied) are, however, a different population. Dropping the missing-value indicators instead, with median or multiple imputation fitted in the training fold, halved PAD-UFES-20's added value at ConvNeXt-T (206.9 to 112.2 and 90.7×10^{-3} nats) but raised HIBA's (3.8 to 12.7); H1's direction held under every encoding. Complete-case retained fractions at ConvNeXt-T range from 0.06 to 0.60.

G. What makes a representation "strong"?

The ladder orders representations by image-only performance but changes architecture, size, and pretraining at once. Four exploratory manipulations separate these factors; their CIs are not corrected for multiplicity.

Fine-tuning. ResNet-18 and ConvNeXt-T were fine-tuned end to end with the same grouped folds and a fixed schedule (AdamW, learning rate 10^{-4}, six epochs, no early stopping on test folds). On HAM10000, fine-tuning ResNet-18 raised image AUROC to 0.92 and lowered added value from 4.1 to 0.3×10^{-3} nats [−3.0, 3.5], a paired drop of 3.8×10^{-3} [1.7, 5.9] (Table VIII). For ConvNeXt-T, whose frozen added value was already small, the drop was not detectable [−1.1, 2.8]. On HIBA and PAD-UFES-20 the drop was significant for both backbones, although on PAD-UFES-20 most of the value remained (ConvNeXt-T: 206.4 to 123.9×10^{-3} nats). On ISIC 2019 and MILK10k (dermoscopic; ResNet-18 only), fine-tuning raised image AUROC (0.802 to 0.836; 0.815 to 0.869) but added value fell only slightly (9.7 to 7.1; 14.7 to 13.7×10^{-3} nats), and neither paired difference excluded zero ([−0.2, 5.7]; [−2.4, 4.5]).

Training-set size. On HAM10000's complete cases, ResNet-18 was fine-tuned on 10%, 25%, 50%, and 100% of each training fold's lesions, with two independent

TABLE VII
Complete-case sensitivity analysis: only images with every metadata field recorded (fold assignment 0; one-sided paired grouped-bootstrap p-values, uncorrected). $\hat{I}_A$ and $\hat{\Delta}$ in 10^{-3} nats; ρ at ConvNeXt-T.

Dataset	Kept (dropped)	$\hat{I}_A$	$\hat{\Delta}_{\text{HSV}}$	$\hat{\Delta}_{\text{ConvNeXt}}$	H1 p	H2 p	ρ
HAM10000	9,761 (254)	26.3	18.1	2.6	$\leq$0.001	0.018	0.10
ISIC 2020	32,531 (595)	4.1	3.4	0.8	$\leq$0.001	$\leq$0.001	0.19
ISIC 2019 (non-HAM)	14,156 (1,160)	12.2	13.8	6.1	0.002	0.162	0.50
PAD-UFES-20	1,089 (1,209)	155.4	152.0	73.8	$\leq$0.001	0.520	0.47
HIBA	1,085 (139)	241.0	75.2	14.1	$\leq$0.001	$\leq$0.001	0.06
MSK 2020	1,157 (17)	36.2	33.6	21.8	0.009	0.073	0.60
MILK10k (derm.)	5,086 (31)	80.5	50.7	12.3	$\leq$0.001	$\leq$0.001	0.15
MILK10k (clin.)	5,086 (31)	80.5	57.6	9.8	$\leq$0.001	$\leq$0.001	0.12

TABLE VIII
Exploratory: frozen vs. end-to-end fine-tuned image models (same grouped folds, fold assignment 0, all images; one training run per fold). $\hat{\Delta}$ and the paired difference (frozen − fine-tuned) in 10^{-3} nats with 95% grouped-bootstrap CI. PAD-UFES-20 includes its label-informative missingness.

Dataset	Backbone	AUROC frz./FT	$\hat{\Delta}$ frz./FT	Difference
HAM10000	ResNet-18	0.854 / 0.917	4.1 / 0.3	3.8 [1.7, 5.9]
HAM10000	ConvNeXt-T	0.887 / 0.930	1.7 / 0.8	0.9 [−1.1, 2.8]
HIBA	ResNet-18	0.908 / 0.940	18.9 / 1.4	17.5 [6.1, 29.8]
HIBA	ConvNeXt-T	0.928 / 0.951	2.6 / −7.2	9.8 [2.6, 17.4]
PAD-UFES-20	ResNet-18	0.845 / 0.899	234.2 / 172.7	61.5 [43.8, 79.5]
PAD-UFES-20	ConvNeXt-T	0.874 / 0.927	206.4 / 123.9	82.5 [65.5, 99.0]
ISIC 2019 (non-HAM)	ResNet-18	0.802 / 0.836	9.7 / 7.1	2.6 [−0.2, 5.7]
MILK10k (derm.)	ResNet-18	0.815 / 0.869	14.7 / 13.7	1.0 [−2.4, 4.5]

subsamples below 100% (7 runs). Mean added value fell from 8.9 to 5.4, 3.2, and 1.2×10^{-3} nats (100%: 95% CI [−1.3, 3.8]) as image AUROC rose from 0.822 to 0.905 (Fig. 3); across runs, the Spearman correlation between training fraction and added value was −0.97. Between 50% and 100%, image AUROC changed little (0.896 vs. 0.905) while added value kept falling, although the intervals of adjacent fractions overlap; image AUROC is thus an incomplete summary of representation strength. Retraining only the head of frozen features on the same fractions moved added value in the same direction but less (8.4 to 4.2 on HAM10000; 10.2 to 7.0 on ISIC 2019).

Backbone size. Within families (ResNet-18/34/50/101; ConvNeXt-T/S/B), frozen and on complete cases of 3 datasets, none of 6 smallest-vs-largest comparisons of added value was significant, and image AUROC changed by only −0.016 to +0.009.

A larger pretrained model. A frozen ViT-L/16 (304M parameters, weakly supervised SWAG pretraining [26]) changed image AUROC relative to ConvNeXt-T by −0.005 to +0.020 across 8 image sets; added value was slightly lower on ISIC 2020, ISIC 2019 (non-HAM) and not higher anywhere.

Together, these results tie attenuation to how well the image model solves the task, not to the number of parameters applied to frozen features.

H. Erasing what the image encodes about age

Proposition 1 predicts that if a variable adds little because the representation already encodes it, removing that encoding should restore its value. We tested this with linear concept erasure [16]: in each training fold, the closed-form projection that makes age linearly undecodable was fitted to the image features and applied with dose t (exploratory; registered before running). One direction sufficed to make age undecodable, and age's added value then rose monotonically with dose on the three datasets where age was decodable (HAM10000: 2.0 to 13.2; MILK10k: 12.4 to 69.0 $\times 10^{-3}$ nats; Fig. 4, Table IX). The rise is not a generic effect of a weaker image model: removing random subspaces of the same rank changed nothing, and removing random subspaces large enough to cost as much image performance raised age's value far less; the paired difference from that matched control excluded zero in 6 of 8 dataset–backbone pairs, namely every pair on the three datasets where age was decodable (on MILK10k the matched control cost somewhat less image performance than erasure: log-loss 0.50 vs. 0.52, ConvNeXt-T). On PAD-UFES-20, where age was barely decodable ($R^2 = 0.02$), erasure had no effect. The effect was specific to the erased variable (HAM10000: age +11.2, site +1.5, sex +0.6), and erasing age cost image performance (log-loss 0.232 to 0.246), so the image models use age-related features for diagnosis. Sex was decodable but carried no added value, and erasing it restored none. These results show the substitution term of Proposition 1 at work for age; they concern linear information only and do not show that training learns age in order to diagnose.

I. Where do metadata help?

Pre-registered before analysis, we built for each training fold a nearest-neighbour graph of the image representation and asked whether per-image benefit is larger where a lesion's neighbours have discordant training labels (H1), where the image model is uncertain (H2), and whether graph statistics improve held-out prediction of benefit beyond the fitted models (H4; Appendix G). At ConvNeXt-T, H2 held in 7 of 7 dataset units, as the bound $u(s) \leq H(Y \mid S = s)$ requires; H1 held in 4 (registered threshold: 5) and H4 in 0. Post hoc, the fusion model's label-free expected gain K ranked per-image benefit best in 6 of 8 image sets, and the quarter of images with the largest K carried 70%–105% of the total benefit (HIBA

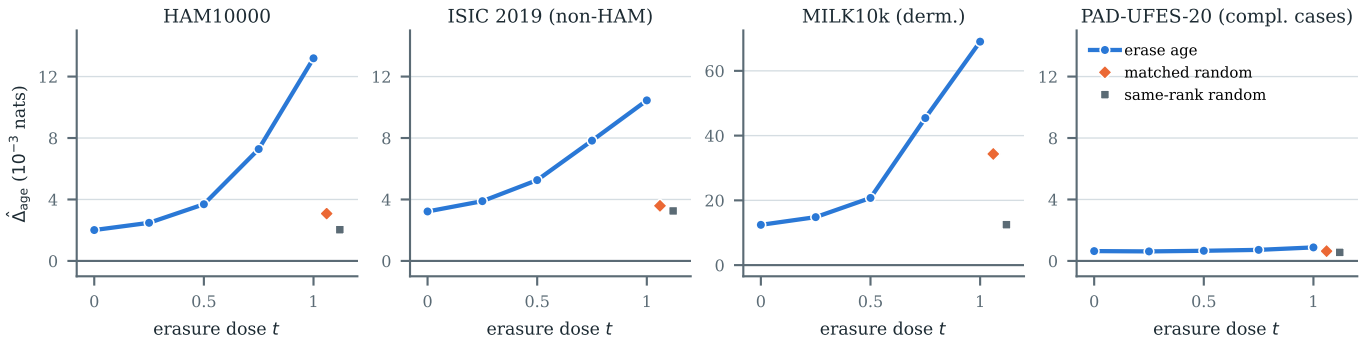


Fig. 4. Exploratory concept erasure (ConvNeXt-T): added value of age as the linear trace of age is removed from the image representation (dose t; $t = 0$: unchanged features). Diamonds and squares at $t = 1$: random subspaces whose removal costs as much image performance (matched) or has the same rank (same-rank); means over five draws. MILK10k is on its own vertical scale; the other panels share one. Strictly cross-fitted estimator, fold assignment 0.

TABLE IX

Exploratory concept erasure (E45/E45b): added value of age (10^{-3} nats) after removing the linear trace of age from the image representation (LEACE, fitted per training fold) at dose t, compared with removing random subspaces. R^2: out-of-fold linear decodability of age before erasure (after erasure: 0.00 throughout). Same rank: mean over 5 random one-dimensional subspaces. Matched: mean over 5 random subspaces whose rank (in brackets) raises image-only log-loss about as much as erasing age. Last column: paired difference, erasure minus matched, with 95% grouped-bootstrap CI. Strictly cross-fitted estimator, fold assignment 0. ‡Complete cases.

Dataset	Backbone	R^2_{age}	$t = 0$	$t = 0.25/0.5/0.75/1$	Same rank	Matched (rank)	Erasure − matched
HAM10000	ConvNeXt-T	0.37	2.0	2.5 / 3.7 / 7.3 / 13.2	2.0	3.1 (64)	10.1 [7.2, 12.9]
	ResNet-18	0.26	2.6	2.9 / 3.9 / 7.2 / 14.2	2.5	3.7 (64)	10.4 [7.6, 13.2]
ISIC 2019	ConvNeXt-T	0.35	3.2	3.9 / 5.3 / 7.8 / 10.4	3.3	3.6 (32)	6.9 [4.6, 9.4]
	ResNet-18	0.29	3.7	4.2 / 5.4 / 7.8 / 10.7	3.7	4.2 (32)	6.4 [4.3, 8.8]
MILK10k (derm.)	ConvNeXt-T	0.31	12.4	14.8 / 20.8 / 45.4 / 69.0	12.5	34.3 (256)	34.7 [28.4, 41.0]
	ResNet-18	0.25	17.5	20.2 / 25.5 / 44.6 / 69.5	17.5	47.9 (256)	21.6 [16.7, 27.1]
PAD-UFES-20‡	ConvNeXt-T	0.02	0.6	0.6 / 0.7 / 0.7 / 0.9	0.6	0.6 (1)	0.2 [−2.5, 3.3]
	ResNet-18	0.01	1.7	2.3 / 2.8 / 3.2 / 3.5	1.7	1.7 (1)	1.8 [−0.1, 3.9]

excluded: total near zero), although K overstated the level of benefit, by up to 9-fold. On ConvNeXt-T, benefit showed no local organisation on the graph beyond label and image uncertainty (normalised Dirichlet energy at most 1% below a stratified permutation null, significant in 2 of 8 sets; colour histograms: 6). In the decomposition $\Delta = H\eta$, falling image uncertainty accounted for only 0.09–0.33 of the log-attenuation from colour histograms to ConvNeXt-T on seven sets (bootstrap upper bounds $\leq$ 0.58), against 0.61 [0.55, 0.66] on PAD-UFES-20 (exploratory): strong representations attenuate metadata mainly by leaving uncertainty that metadata no longer resolve, not by resolving more cases.

VI. Discussion and Limitations

What the results mean. On every independent dataset, metadata add clearly more to a colour-histogram model than to frozen deep representations, and better task-specific image models (fine-tuned, or trained on more data) leave less for the metadata to add, although on ISIC 2019 and MILK10k the drop after fine-tuning was small and not significant. Attenuation between frozen deep representations is smaller and inconsistent (H2: 3 of 6), as expected if those representations differ little in task performance. How much value remains at a strong representation depends on the dataset and its label definition, from little (HAM10000; HIBA, depending on weighting and missing-value encoding) to about half (PAD-UFES-20, MSK 2020, ISIC 2019 for melanoma), and not on imaging modality: the pre-registered prediction that clinical images leave more value than dermoscopy of the same lesions failed. Concept erasure shows the redundancy at work for age: the value age loses to strong representations returns when their age information is removed. The frequently reported benefit of metadata [1], [2] is therefore partly a statement about weak image representations, and the size of the part that is not differs severalfold between datasets.

Which variables keep their value? In an exploratory analysis over 95 dataset–representation–variable combinations (of 160; the rest had no significant standalone value), variables that the image predicts well within a diagnostic class kept a smaller share of their value (Spearman −0.75; partial correlation adjusting for standalone value and image AUROC −0.59, dataset-clustered 95% CI [−0.72, −0.40]; between −0.41 and −0.75 within each representation). This is the pattern Proposition 1 predicts, but it is correlational, and it holds only among variables with demonstrable standalone value: there it survives a mixed model with dataset effects (coefficient −0.49 [−0.67,

−0.31]), whereas over all 160 combinations it weakens (partial −0.28; mixed −0.10 [−0.25, 0.05]). A transparent rule fitted to these quantities did not predict a variable's added value in held-out datasets better than predicting zero (mean absolute error 0.0101 vs. 0.0092 nats); the simple product of standalone value and a typical retained fraction did best (0.0082). As a yes/no decision (collect if the predicted value is at least 10^{-3} nats), that simple product was right in 0.80 of cases (precision 0.81, recall 0.83; Appendix E). With six independent datasets, a reliable quantitative rule is beyond reach.

Under dataset shift. When the image and fusion models were fitted on one dataset and applied to another (harmonised age, sex, and site; 26 directed pairs), metadata improved log-loss significantly in 14 pairs and worsened it in 7, from −0.147 to 0.063 nats. The largest harms occurred between datasets with very different prevalences and label definitions, so these results describe the joint shift rather than a property of metadata. Even within one dataset, added value learned on some sources did not transfer: holding out one HAM10000 source at a time (all images, strictly cross-fitted, ConvNeXt-T, 10^{-3} nats): rosendahl −6.5 [−16.8, 3.6]; vidir_modern −14.4 [−32.1, 0.9]; vidir_molemax 7.6 [6.1, 9.1]; vienna_dias 18.0 [3.2, 30.6]; ISIC 2019, BCN to MSK −10.5.

Practical implication. Metadata fusion should be evaluated against the strongest available image model, ideally one fine-tuned on the target data, and the retained fraction ρ reported alongside the gain, so that a benefit that merely compensates for a weak backbone is not mistaken for complementary information. Because benefit is concentrated, the fusion model's expected gain can rank the lesions whose metadata matter most, although it overstates the size of the gain.

Limitations. (i) Fine-tuning was examined on five datasets (two with ResNet-18 only) with one training run per fold and a fixed schedule; the drop in added value was significant on three, and fine-tuned scores were not re-estimated with the strictly cross-fitted second stage. (ii) $\hat{\Delta}$ estimates the reduction achievable by a logistic second stage on a one-dimensional score, not $I(Y; A \mid S)$ itself. (iii) The frozen ladder is ordered by image-only performance and is not a controlled intervention; Section V-G addresses this only in part. (iv) Both pre-registered mechanism tests failed because their premises (invisible symptoms; a weaker clinical-image representation) did not hold; the mechanism is shown by intervention only for age and only for linearly decodable information. (v) Images come from public archives and mirrors; official metadata were used and counts were checked against the published descriptions. MSK 2020 has no lesion identifier and is grouped per image (an embedding check found no near-duplicate images within it); ISIC 2019 images without one are grouped individually. (vi) Missingness is label-informative in every dataset, severely so in PAD-UFES-20; complete-case analysis removes this leak but analyses a selected subsample, and some exploratory PAD-UFES-20 results (Table VIII) use all images. (vii) All analyses are retrospective; nothing here is a clinical claim, and no causal effect of metadata is estimated. (viii) Bootstrap CIs hold the fitted models fixed and omit training variability [21]: across fold assignments, added value at ConvNeXt-T varied by 22.2×10^{-3} nats on PAD-UFES-20 (CI half-width 27.1) and HIBA's took the values 3.8, 10.0, 9.1. (ix) Saliency analyses (Appendix H) are evidence of spatial correspondence, not of causal reliance [27].

VII. Conclusion

The value of patient metadata to a skin lesion classifier is not fixed: it depends on what the image model already extracts. Proposition 1 makes this precise. Pre-registered experiments on seven datasets show that metadata add far less to deep representations than to colour histograms, a result that survives a strictly cross-fitted estimator, alternative labels, complete cases and missing-data encoding. Interventions show what drives it: fine-tuning and more training data tend to shrink the gain, more parameters on frozen features do not, and erasing what a representation encodes about age restores age's value. Attenuation among frozen deep representations did not replicate consistently, the share of metadata information left non-redundant ranged severalfold across datasets and with the label definition, imaging modality did not explain that range, and the remaining benefit concentrated in lesions the image model found uncertain. Claims that metadata improve a classifier should therefore be made against strong, task-adapted image baselines and accompanied by the retained fraction.

Appendix A
Index of experiments

Table X lists every experiment of this study and where it is reported. Earlier experiments in the same project (E01–E08: an Alzheimer's MRI study; E11: industrial anomaly detection) address unrelated questions and are not part of this paper.

Appendix B
Simulation of the attenuation signature

In a simulation with a known answer (E09), a latent image variable generated the label, and an auxiliary variable acted either through the image, directly on the label, both, or not at all. A ladder of image models was created by adding decreasing noise to the image representation. Added value fell to zero when the auxiliary variable acted through the image, as its recoverability from the image rose, and rose when it acted directly (Table XI); the latter is the non-collapsibility of logistic regression [13] and corresponds to $R(S) < 0$ in Proposition 1. The mixed case fell to a non-zero floor.

Appendix C
Further manipulations of the image representation

Tables XII–XIV give the full results summarised in Section V-G. Table XV reports a resolution ladder (E19): with the architecture fixed, images were downsampled before embedding. The prediction that added value falls monotonically with resolution held in 3 of the 7 image sets analysed (PAD-UFES-20, MILK10k (derm.), MILK10k (clin.); the PAD-UFES-20 analysis uses all images and therefore includes its missingness leak), and image AUROC hardly changed (on HAM10000 it is 0.88 even at 16×16 pixels), so resolution is a weak handle on representation strength for these tasks.

Appendix D
Per-variable floors

Single-variable added values (Table XVI) show that every variable with detectable value at the colour-histogram rung attenuates on HAM10000 and ISIC 2020 (on ISIC 2020, sex and site carry no detectable value at all), whereas on ISIC 2019 anatomical site keeps most of its value. This is consistent with ISIC 2019's large non-redundant share (Table VI) and with BCN20000's deliberate inclusion of hard-to-diagnose sites [5].

Appendix E
Exploratory analyses

Recoverability (Fig. 5). The negative association reported in Section VI does not depend on the colour histogram: restricted to the deep representations, the within-dataset Spearman correlation is negative in 8 of 8 image sets (from -0.40 to -0.95; as few as 4 points in some sets). The rule excluding rows with non-significant standalone value was fixed in the analysis code before the run but was not stated in the dated registry.

Permutation control (Table XVII). Permuting whole metadata records between grouping units destroys their link to the image and label but keeps their joint distribution; permuted metadata never matched the real metadata's added value and always had negative added value on average, as expected from fitting extra parameters to noise.

Collection rule (Table XVIII). A ridge model on six cheap indicators was worse than predicting zero; the product of standalone value and the median retained share of the training datasets was the best predictor and made a useful collect/skip decision.

Dataset shift (Table XIX). Under shift, metadata fitted on the source frequently harmed the target, most strongly between datasets with very different prevalences. Within HAM10000, holding out one image source at a time (10^{-3} nats, 95% CI): rosendahl -5.9 [-13.9, 2.0]; vidir_modern 6.3 [1.8, 11.2]; vidir_molemax 7.4 [6.2, 8.6]; vienna_dias 16.2 [5.2, 28.0].

Appendix F
Robustness of the replication

Table XX re-tests H1 and H2 under the conditions of the audit. Alternative labels: malignant vs. benign ({mel, bcc} on HAM10000; {MEL, BCC, SCC} on ISIC 2019) and melanoma vs. rest (PAD-UFES-20, HIBA, MSK 2020, MILK10k). H1 held under the alternative label in 5 of 6 independent datasets and on complete cases in 8 of 8 image sets; H2 held on complete cases in 6 of 8 (uncorrected).

Appendix G
Where metadata help: details

For each outer training fold, image features were standardised with training-fold statistics and L_2-normalised; each training image was linked to its 20 nearest training images from other grouping units (cosine distance, Gaussian weights with bandwidth equal to the median 20th-neighbour distance, union-symmetrised), and each test image to its 20 nearest training images, so no test label entered any statistic. Local label entropy uses the neighbours' training labels; graph-local benefit averages the neighbours' inner out-of-fold benefit computed within the training fold. Results were the same with 10 or 50 neighbours and with binary weights. H4 compared ridge regressions with and without graph statistics, both including K, image entropy, margin, feature norm and local density; both had low held-out R^2, because their training targets came from models fitted on less data, so the post hoc rank correlations in Table XXI are the more informative comparison. Per-sample prediction of a feature's incremental value is established in dynamic feature selection [15]; these analyses ask only where the benefit lies.

Appendix H
Interpretability

Interpretability test. For a linear head w on global-average-pooled features $f = \frac{1}{|\Omega|}\sum_{u\in\Omega} F(u)$, the class activation map $c(u) = \sum_j w_j F_j(u)$ [28] decomposes the logit exactly. For a non-negative evidence map e and a binary mask m on a periodic grid Ω, the mass inside the mask after a shift by τ is

$$M(\tau) = \frac{\sum_u e(u)\, m(u-\tau)}{\sum_u e(u)} = \frac{(e \star m)(\tau)}{\|e\|_1}, \qquad (4)$$

a circular cross-correlation. Hence $\mathcal{F}^{-1}[\mathcal{F}(e)\overline{\mathcal{F}(m)}]$ yields $M(\tau)$ for all $|\Omega|$ shifts in $O(|\Omega|\log|\Omega|)$, and $|\{\tau : M(\tau) \geq M(0)\}|/|\Omega|$ is an exact p-value under the hypothesis that the mask's position is uniform over circular translations. Unlike pixel permutation, the shift null preserves the spatial autocorrelation of e.

For each backbone we use the exact class activation map of the out-of-fold head. For ConvNeXt-T, whose pooled features pass through a LayerNorm before the head, the map is exact up to a positive per-image factor: with per-image LayerNorm scale σ and affine weights γ, the logit

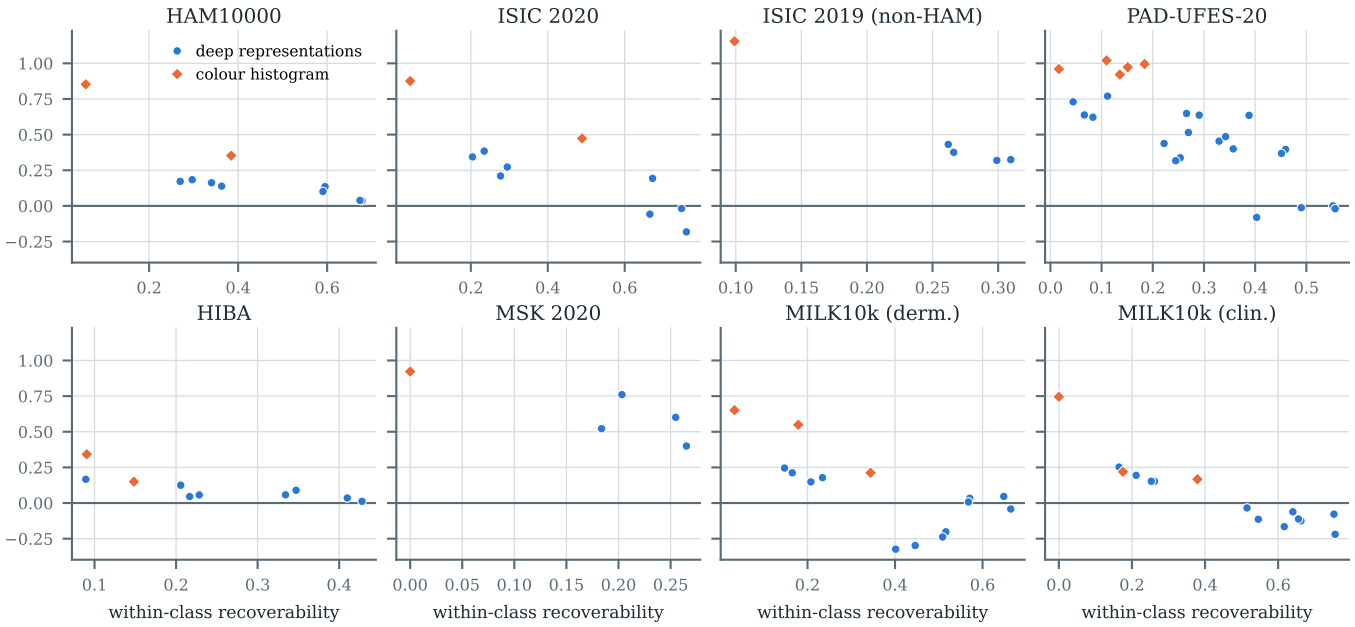


Fig. 5. Exploratory (E30): retained share of each variable's added value ($\hat{\Delta}_{\text{var}}/\hat{I}_{\text{var}}$) against its within-class recoverability from the image representation, for the 95 combinations where the variable's standalone value was significant. Orange: colour histogram; blue: deep representations.

equals a constant plus $\sigma^{-1}|\Omega|^{-1}\sum_u \sum_j \tilde{w}_j\big(F_j(u)-\bar{F}(u)\big)$ with $\tilde{w}_j = w_j\gamma_j/s_j$ and $\bar{F}(u)$ the channel mean. The overlap statistics are invariant to that factor, and the reconstructed logits matched the head's logits with correlation 1.000 for every backbone.

Mapping the predicted class's evidence produced an apparent link between errors and off-lesion attention ($p < 10^{-49}$; Table XXIII). This is an artefact: for "not melanoma" predictions, the evidence for that class lies in the surrounding skin. With the melanoma-evidence map used for every image (Table XXIV), melanoma evidence is concentrated on the lesion in 84%–91% of images across the three backbones, a criterion that a centred map would also meet. Missed melanomas show no less on-lesion evidence; instead they are consistently smaller lesions ($p \leq 5 \times 10^{-12}$ for every backbone). Pixel permutation declares 82%–90% of maps significant, against 36%–56% under the circular-shift null, consistent with the inflation predicted by its disregard of spatial autocorrelation [18], [19].

The shift null is itself invalid here. It is exact only if, given the map, the mask's position is uniform over circular translations. Dermoscopic lesions are centred, so any centred map passes it. A fixed centred Gaussian, which carries no image information, was significant for 88% of images under the shift null and for 5% under a mask-swap null that compares each image's own mask with the masks of 199 images of similar lesion area (E48; Table XXII). Under the swap null, 10%–19% of real maps were significant (shift null on the same images: 37%–57%): some lesion-specific evidence exists, but most of the shift-null significance reflects centring. Preserving spatial autocorrelation is necessary for a valid spatial null but not sufficient; the transformation must also leave the joint distribution of map and mask unchanged under the null.

TABLE X
All experiments of this study. Pre-reg.: hypotheses, tests and correction fixed before modelling. Exploratory: registered or described before running where stated, but not confirmatory.

ID	Question	Status	Reported in	Outcome
E09	Does the slope of added value along a ladder separate mediated from direct information?	Simulation	App. B	Yes: mediated falls, direct rises
E10	Does added value attenuate along the ladder on HAM10000?	Pre-reg.	Sec. V-A, Tab. II	3 of 4 predictions held; no non-zero floor
E10a	Which metadata fields leak the label?	Audit	Sec. IV	dx_type, biopsed excluded; image source is a shortcut
E10b	Is attenuation a source shortcut? Paired tests	Exploratory	Tab. IV	Holds with source adjustment
E10c, E20	Which variables attenuate?	Exploratory	Sec. V-A, Tab. XVI	All attenuate together; ISIC 2019 floor carried by site
E10d	Effect in AUROC and Brier units	Exploratory	Tab. III	ΔAUROC 0.043 $\rightarrow$ 0.003
E12, E12b	Does evidence lie on the lesion? (ResNet-50)	Exploratory	Tab. XXIII	Yes; sign-convention pitfall found
E13 (wave 1)	Replication on ISIC 2020, ISIC 2019, PAD-UFES-20	Pre-reg.	Tab. V	H1 3/3, H2 1/3; PAD mechanism M1–M3 failed
E13/E22 (wave 2)	Replication on HIBA, MSK 2020, MILK10k; P1, P2	Pre-reg.	Tab. V, Sec. V-D	H1 4/4, H2 3/4; P1 failed; P2 held
E14, E14b	Frozen vs. fine-tuned	Exploratory	Tab. VIII	Added value lower after fine-tuning in all 8 pairs; significant on HAM10000 (ResNet-18), HIBA, PAD-UFES-20
E31c, E31d	Fine-tuning on 10–100% of the data	Exploratory*	Fig. 3	Added value falls monotonically with training data
E15	Interpretability across backbones	Exploratory	App. H	Misses are small lesions; on-lesion evidence beyond centring in 10%–19% of maps (E48)
E16	Are PAD-UFES-20 symptoms visible?	Exploratory	Sec. V-C	Yes, partly, within class
E17	Retained fraction ρ	Exploratory	Tab. VI, Fig. 2	Dataset-specific floors
E18	Does a 304M-parameter ViT attenuate further?	Exploratory	Tab. XIV	No consistent change
E19	Does lower resolution restore added value?	Exploratory	Tab. XV	Weak manipulation; mixed
E21	Complete-case re-analysis	Exploratory	Tab. VII	H1 8/8, H2 5/8 (uncorrected)
E30	Does recoverability predict what is retained?	Exploratory*	Sec. VI, Fig. 5	Negative association within every representation
E31a, E31b	Backbone size; data size with frozen features	Exploratory*	Sec. V-G, Tabs. XII, XIII	Data and fine-tuning matter; size does not
E32	Permutation control	Exploratory*	Sec. V-B, Tab. XVII	Real metadata beat all permutations
E33	Cross-dataset shift	Exploratory*	Sec. VI, Tab. XIX	Mixed; confounded with prevalence
E34	A rule for which metadata to collect	Exploratory*	Sec. VI, Tab. XVIII	Fitted rule failed; simple baseline best
E40	Does a strictly cross-fitted second stage change the results?	Exploratory*	Sec. IV, App. F	Changes $\leq$0.42 CI half-widths; H1 unchanged; ISIC 2020 H2 now holds
E41	Alternative labels; complete cases, corrected	Exploratory*	Tab. XX	H1 robust except MILK10k melanoma; ρ label-dependent
E42	Does graph ambiguity locate or predict benefit?	Pre-reg.†	Sec. V-I, Tab. XXI	H1 4/7 (failed), H2 7/7, H4 0/7 (failed)
E44	Is benefit smooth on the graph?	Exploratory*	Sec. V-I, Tab. XXI	Not beyond label and uncertainty (ConvNeXt-T)
E45, E45b	Does erasing age from the representation restore its value?	Exploratory*	Sec. V-H, Tab. IX	Yes where age is decodable; beyond matched random erasure in 6/8 pairs
E46	E30 without its filter	Exploratory*	Sec. VI	Holds only for variables with standalone value
E47	Missing-data encoding; source-held-out evaluation	Exploratory*	Sec. V-F, Sec. VI	Encoding changes added value up to threefold; H1 direction unchanged
E48	Is the shift null valid for centred lesions?	Exploratory*	App. H, Tab. XXII	No: a centred Gaussian passes in 88% of images
E49	Uncertainty for the $H\eta$ decomposition	Exploratory*	Sec. V-I	Image uncertainty explains 0.09–0.33 of attenuation (PAD-UFES-20: 0.61)

*Registered in a dated document before implementation.
†Hypotheses and decision rules fixed in a dated registry before implementation.

TABLE XI
Simulation (E09; $n = 3{,}000$, 10 repetitions, 6 image-noise levels). Mean added value (10^{-3} nats) and recoverability (R^2) of the auxiliary variable from the image at the weakest and strongest image model.

	$\hat{\Delta}$		Recoverability	
Auxiliary effect	weakest	strongest	weakest	strongest
Through the image	2.4	−0.7	0.10	0.57
Direct	29.4	43.6	−0.02	−0.02
Both	8.2	1.6	0.11	0.58
None	−0.2	0.0	−0.02	−0.02

TABLE XII
Backbone size within families (E31a, exploratory; frozen, complete cases, fold assignment 0). Each cell: image AUROC / $\hat{\Delta}$ (10^{-3} nats). No smallest-vs-largest difference in $\hat{\Delta}$ was significant (paired grouped bootstrap).

Dataset	ResNet-18	ResNet-34	ResNet-50	ResNet-101	ConvNeXt-T	ConvNeXt-S	ConvNeXt-B
HAM10000	0.859 / 4.2	0.854 / 4.8	0.851 / 4.0	0.843 / 3.7	0.891 / 2.6	0.875 / 3.2	0.891 / 1.6
PAD-UFES-20	0.779 / 73.4	0.773 / 87.6	0.771 / 83.1	0.771 / 87.9	0.776 / 73.8	0.779 / 68.3	0.771 / 77.8
ISIC 2019 (non-HAM)	0.802 / 7.0	0.807 / 7.2	0.803 / 6.7	0.812 / 6.1	0.825 / 6.1	0.820 / 6.2	0.827 / 5.6

TABLE XIII
Training-set size with frozen features (E31b, exploratory; complete cases). The image head sees 10–100% of each training fold's groups; the metadata stage uses all folds. Each cell: image AUROC / $\hat{\Delta}$ (10^{-3} nats), mean of two subsamples below 100%.

Dataset	Features	10%	25%	50%	100%
HAM10000	ResNet-18	0.798 / 8.4	0.828 / 7.0	0.842 / 4.9	0.859 / 4.2
	ConvNeXt-T	0.810 / 7.4	0.852 / 3.5	0.877 / 3.8	0.891 / 2.6
ISIC 2019 (non-HAM)	ResNet-18	0.747 / 10.2	0.775 / 7.0	0.793 / 7.0	0.802 / 7.0
	ConvNeXt-T	0.767 / 9.0	0.797 / 6.8	0.811 / 5.8	0.825 / 6.1

TABLE XIV
A larger pretrained representation (E18, exploratory; frozen, all images, fold assignment 0): ConvNeXt-T vs. ViT-L/16 SWAG [26]. $\hat{\Delta}$ and the paired difference (ConvNeXt − ViT-L) in 10^{-3} nats.

	Image AUROC		$\hat{\Delta}$		
Dataset	CNX-T	ViT-L	CNX-T	ViT-L	Difference
HAM10000	0.887	0.896	1.7	3.4	−1.6 [−3.6, 0.4]
ISIC 2020	0.859	0.879	1.0	0.5	0.5 [0.2, 0.8]
ISIC 2019 (non-HAM)	0.823	0.828	9.1	6.7	2.4 [0.2, 4.7]
PAD-UFES-20	0.874	0.886	206.4	189.7	16.8 [−1.5, 34.7]
HIBA	0.928	0.936	2.6	−0.0	2.6 [−6.1, 10.8]
MSK 2020	0.727	0.730	18.0	18.7	−0.7 [−6.0, 4.6]
MILK10k (derm.)	0.845	0.840	8.7	8.2	0.5 [−1.9, 3.0]
MILK10k (clin.)	0.842	0.840	5.6	5.9	−0.2 [−2.0, 1.5]

TABLE XV

Resolution ladder (E19, exploratory): frozen ConvNeXt-T on images downsampled to $r \times r$ pixels and resized back. Image AUROC at the extremes; $\hat{\Delta}$ (10^{-3} nats) at each resolution; Spearman correlation of resolution with $\hat{\Delta}$ (prediction: -1). The manipulation is weak: image AUROC barely depends on resolution. All images (PAD-UFES-20 includes its missingness leak).

	Image AUROC		$\hat{\Delta}$ at resolution					
Dataset	16 px	224 px	16	32	64	112	224	Spearman
HAM10000	0.880	0.887	4.1	3.2	3.0	3.4	1.7	−0.7
ISIC 2020	0.850	0.859	1.3	1.2	0.8	0.9	1.0	−0.6
ISIC 2019 (non-HAM)	0.804	0.823	12.2	9.0	8.7	8.4	9.1	−0.4
PAD-UFES-20	0.849	0.874	246.5	228.7	224.9	212.6	206.4	−1.0
HIBA	0.924	0.928	5.7	6.9	9.4	10.3	2.6	0.0
MILK10k (derm.)	0.827	0.845	12.1	10.3	11.0	9.9	8.7	−0.9
MILK10k (clin.)	0.835	0.842	10.6	10.2	8.5	6.8	5.6	−1.0

TABLE XVI

Single-variable added value (E20, exploratory; 10^{-3} nats, 95% CI, all images). Retained: ConvNeXt-T / HSV, shown only where the HSV value is significant. PAD-UFES-20 is omitted: its per-variable results were driven by label-informative missingness (Section V-F).

Dataset	Variable	HSV	ConvNeXt-T	Retained
HAM10000	age	14.7 [9.9, 19.6]	2.0 [−1.0, 4.5]	0.13
	sex	2.5 [0.8, 4.2]	0.4 [0.0, 0.8]	0.14
	site	5.7 [2.2, 9.3]	1.1 [−1.6, 3.7]	0.19
ISIC 2020	age	3.3 [2.1, 4.4]	1.1 [0.5, 1.7]	0.32
	sex	0.2 [−0.0, 0.5]	0.1 [−0.1, 0.3]	–
	site	0.3 [−0.2, 0.8]	−0.1 [−0.4, 0.2]	–
ISIC 2019 (non-HAM)	age	11.2 [5.7, 16.5]	3.2 [0.6, 5.7]	0.28
	sex	−0.8 [−1.6, −0.2]	−0.2 [−0.8, 0.3]	–
	site	7.1 [2.2, 12.6]	5.4 [0.1, 10.7]	0.76

TABLE XVII

Permutation control (E32, exploratory; ConvNeXt-T, complete cases): $\hat{\Delta}$ (10^{-3} nats) with the real metadata and the mean and maximum over 50 permutations of whole metadata records between grouping units. Last column: % of records with an implausible combination (e.g. sex-specific site), real / permuted, all images.

Dataset	Real	Perm. mean	Perm. max	Implausible (%)
HAM10000	2.6	−2.3	0.0	0.2 / 0.6
ISIC 2020	0.8	−0.2	0.2	0.1 / 0.1
ISIC 2019 (non-HAM)	6.1	−2.2	0.2	0.3 / 2.2
PAD-UFES-20	73.8	−28.1	−8.0	0.2 / 2.3
HIBA	14.1	−8.2	3.0	0.2 / 1.5
MSK 2020	21.7	−3.2	2.3	0.0 / 0.3
MILK10k (derm.)	12.3	−1.0	0.2	0.2 / 2.6
MILK10k (clin.)	9.8	−1.1	0.3	0.2 / 2.6

TABLE XVIII

Predicting a variable's added value in a held-out dataset (E34, exploratory; leave-one-dataset-out, MILK10k held out as one; 160 dataset–representation–variable rows). MAE in 10^{-3} nats. Decision rule fixed in advance: collect if predicted $\hat{\Delta}_{\text{var}} \geq 10^{-3}$ nats.

Predictor	MAE	Spearman	Precision	Recall	Accuracy
Predict zero	9.2	–	–	0.00	0.45
$\hat{I}_{\text{var}}\times$ median retained	8.2	0.74	0.81	0.83	0.80
Ridge (6 features)	10.1	0.64	0.61	0.93	0.63

TABLE XIX

Metadata added value under dataset shift (E33, exploratory; 10^{-3} nats). Image head and fusion model fitted on the source only (frozen ConvNeXt-T; harmonised age, sex, coarse site; complete cases) and applied unchanged to the target. $^{+}/^{-}$: 95% target-group bootstrap CI above/below 0. Diagonal (italic): the target's own cross-validated value with the same harmonised metadata. Melanoma and malignancy datasets are paired only within their label family. Differences in prevalence and label definition between datasets are not separated from metadata effects.

Source ↓ / target →	HAM10000	ISIC 2020	ISIC 2019 (non-HAM)
HAM10000	2.1	1.1^{+}	-26.8^{-}
ISIC 2020	15.9^{+}	0.8	11.3^{+}
ISIC 2019 (non-HAM)	9.7^{+}	6.7^{+}	6.0

Source ↓ / target →	PAD-UFES-20	HIBA	MSK 2020	MILK10k (derm.)	MILK10k (clin.)
PAD-UFES-20	2.9	−6.9	30.1^{+}	−2.8	-15.7^{-}
HIBA	-20.2^{-}	9.7	58.5^{+}	28.1^{+}	55.0^{+}
MSK 2020	-64.2^{-}	6.1	21.7	-146.6^{-}	-115.3^{-}
MILK10k (derm.)	-21.1^{-}	57.2^{+}	−3.5	12.3	62.7^{+}
MILK10k (clin.)	−7.7	56.0^{+}	20.6^{+}	25.9^{+}	9.8

TABLE XX

Robustness of H1 and H2 (✓: holds; ×: fails; –: not derivable). Registered and strictly cross-fitted rows: Holm-adjusted within the registered waves (HAM10000 unadjusted). Alternative label: malignant vs. benign for HAM10000 and ISIC 2019, melanoma vs. rest elsewhere; ISIC 2020 has no distinct alternative. Alternative-label, complete-case and encoding rows use the strictly cross-fitted estimator and uncorrected one-sided tests ($p < 0.05$) or point estimates as stated; fold assignment 0 unless stated.

Condition	HAM	ISIC20	ISIC19	PAD	HIBA	MSK	MILK-d	MILK-c
H1: $\Delta_{\text{HSV}} > \Delta_{\text{ConvNeXt}}$								
Original estimator (registered)	✓	✓	✓	✓	✓	✓	✓	✓
Strictly cross-fitted	✓	✓	✓	✓	✓	✓	✓	✓
Fold assignments 0/1/2 (point > 0)	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
Group-weighted (point > 0)	✓	✓	✓	✓	✓	✓	✓	✓
Alternative label	✓	–	✓	✓	✓	✓	×	×
Complete cases	✓	✓	✓	✓	✓	✓	✓	✓
No missing indicators (point > 0)	✓	✓	✓	✓	✓	✓	✓	✓
H2: $\Delta_{\text{ResNet-18}} > \Delta_{\text{ConvNeXt}}$								
Original estimator (registered)	✓	×	×	✓	✓	×	✓	✓
Strictly cross-fitted	✓	✓	×	✓	✓	×	✓	✓
Fold assignments 0/1/2 (point > 0)	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3
Group-weighted (point > 0)	✓	✓	×	✓	✓	✓	✓	✓
Alternative label	✓	–	✓	✓	×	×	×	×
Complete cases	✓	✓	×	×	✓	✓	✓	✓
No missing indicators (point > 0)	–	–	–	–	–	–	–	–

TABLE XXI

Where metadata help (ConvNeXt-T unless stated; strictly cross-fitted, fold assignment 0). ρ_{LE}, ρ_h: Spearman correlation of realised per-image benefit with local label entropy on the inductive graph (pre-registered H1, 95% grouped CI) and with the image model's predictive entropy (H2). H4 p: one-sided, unadjusted, graph statistics vs. model-based baselines. ρ_K, ρ_G: post hoc rank correlation of benefit with the fusion model's expected gain K_i and with graph-local benefit. Top 25%: share of total benefit captured by the quarter of images with the largest K_i (HIBA omitted: total benefit near zero). E: normalised Dirichlet energy of benefit relative to the label-by-uncertainty stratified null (HSV, ConvNeXt-T). s_H: share of the HSV→ConvNeXt-T log-attenuation due to image uncertainty (exploratory).

Image set	ρ_{LE}	ρ_h	H4 p	ρ_K	ρ_G	Top 25%	E_{HSV}	E_{CNX}	s_H
HAM10000	0.03 [−0.00, 0.05]	0.09	0.822	0.13	0.11	90%	0.986	0.997	0.13
ISIC 2020	0.11 [0.08, 0.13]	0.16	0.014	0.15	0.19	97%	0.986	0.999	0.09
ISIC 2019 (non-HAM)	0.10 [0.07, 0.13]	0.17	0.281	0.21	0.15	105%	0.976	0.989	0.33
PAD-UFES-20	0.26 [0.22, 0.30]	0.53	0.943	0.69	0.13	70%	1.003	1.002	0.61
HIBA	0.02 [−0.04, 0.09]	0.13	0.988	0.15	−0.02	–	0.979	1.005	0.13
MSK 2020	−0.00 [−0.06, 0.06]	0.09	0.011	0.18	0.15	86%	1.002	0.994	0.12
MILK10k (derm.)	0.12 [0.09, 0.15]	0.17	0.003	0.18	0.12	82%	0.989	0.995	0.13
MILK10k (clin.)	0.16 [0.13, 0.18]	0.17	0.373	0.16	0.12	76%	0.973	0.997	0.11

TABLE XXII
Share of images whose melanoma-evidence map overlaps its lesion mask significantly ($p \leq 0.05$) under the circular-shift null and under the mask-swap null (masks of 199 other images in the same lesion-area decile), for real class-activation maps and for a fixed centred Gaussian map that carries no image information (E48; HAM10000).

		Real maps		Centred Gaussian	
Backbone	Images	Shift	Swap	Shift	Swap
ResNet-18	2,225	55%	16%	88%	5%
ResNet-50	2,225	37%	10%	88%	5%
ConvNeXt-T	2,225	57%	19%	88%	5%

TABLE XXIII
Class-activation evidence vs. expert lesion masks (HAM10000, ResNet-50, 2225 images). Enrichment: share of positive evidence inside the lesion divided by the lesion's area share. Significance at $p \leq 0.05$.

Evidence map	Enrichment > 1	Shift null	Pixel perm.
Predicted class	32%	10%	30%
Melanoma (all images)	84%	37%	82%

TABLE XXIV
Melanoma evidence (exact CAM) vs. expert lesion masks across backbones (HAM10000; all melanomas plus a matched sample of other lesions). Significance at $p \leq 0.05$: exact circular-shift null vs. pixel permutation. Last column: median lesion area fraction of detected / missed melanomas.

Backbone	Images	Enrich.> 1	Shift null	Pixel perm.	Area det./miss.
ResNet-18	2,222	90%	53%	89%	0.49 / 0.37
ResNet-50	2,222	84%	36%	82%	0.52 / 0.37
ConvNeXt-T	2,222	91%	56%	90%	0.49 / 0.33

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